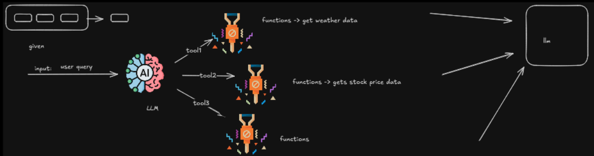
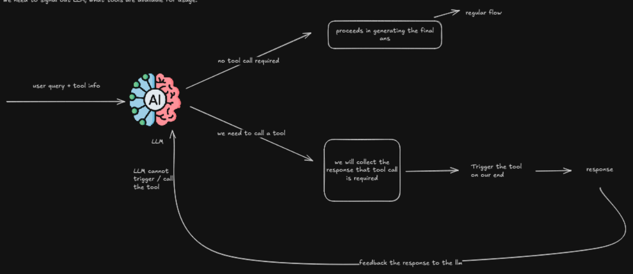

given
input: user query
LLM
tool1
tool2
tool3
functions -> get weather data
functions -> gets stock price data
functions
llm

We need to signal out LLM, what tools are available for usage.


user query + tool info

LLM

no tool call required

proceeds in generating the final ans

regular flow

we need to call a tool

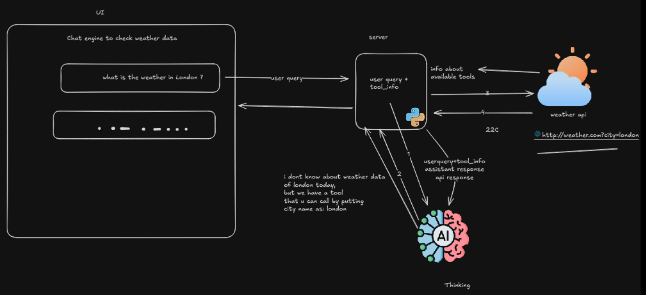
we will collect the response that tool call is required

Trigger the tool on our end

response

feedback the response to the llm

LLM cannot trigger / call the tool


UI

Chat engine to check weather data

what is the weather in London ?

user query

server

user query + tool_info

info about available tools

weather api

22C

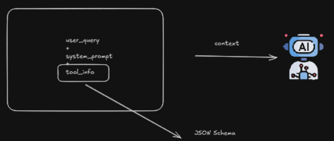
http://weather.com?city=london

userquery+tool_info
assistant response
api response

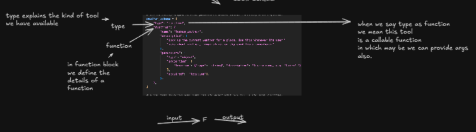
i dont know about weather data
of london today,
but we have a tool
that u can call by putting
city name as: london

Thinking

LLM will tell which tool to call and with what parameters to call.


user_query
+
system_prompt
+
tool_info

context


JSON Schema

type explains the kind of tool we have available

in function block we define the details of a function

when we say type as function
we mean this tool
is a callable function
in which may be we can provide args
also.

input → F → output

User: Should I bring an umbrella in London ?

↓

API call with tools and query and system prompt -> [schema]

↓

Model reals the full context, and understand that we need tool calls to answer

↓

model returns says tool_Calls are required, (not running your code, just a response)
it will also tell which tool to call and with what parameters:
name: lookup_Weather
argument: '{ location: 'london' }' ← JSON string

↓

we will get this structured response in the code

↓

we make the api call with location name deicded by llm

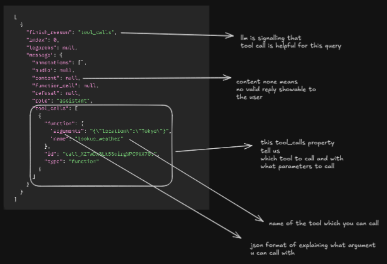
↓

with the response of api call, we send the result and full context back to the llm.

↓

model writes the final answer.

```
[Choice(finish_reason='tool_calls', index=0, logprobs=None, message=ChatCompletionMessage(content=None, refusal=None, role='assistant',
annotations=[], audio=None, function_call=None, tool_calls=[ChatCompletionMessageFunctionToolCall(id='call_XZTwDx8LkBSoirgWPC9UX7Gt',
function=Function(arguments='{"location":"Tokyo"}', name='lookup_weather'), type='function')]))]
```



```
[
  {
    "finish_reason": "tool_calls",
    "index": 0,
    "logprobs": null,
    "message": {
      "annotations": [],
      "audio": null,
      "content": null,
      "function_call": null,
      "refusal": null,
      "role": "assistant",
      "tool_calls": [
        {
          "function": {
            "arguments": "{\"location\":\"Tokyo\"}",
            "name": "lookup_weather"
          },
          "id": "call_XZTwDx8LkBSoirgWPC9UX7Gt",
          "type": "function"
        }
      ]
    }
  }
]
```

llm is signalling that
tool call is helpful for this query

content none means
no valid reply showable to
the user

this tool_calls property
tell us
which tool to call and with
what parameters to call

name of the tool which you can call

json format of explaining what argument
u can call with

system_prompt
user_query
assistant
user_query
assistant